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Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2024

Time: 03 Hours

Code: 101507 ***Subject: Transportation Engineering*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven questions only): **[14]**

(a) The particular places where pedestrians are to cross the pavement are properly marked by the pavement marking known as
(i) stop lines (iii) turn markings
(ii) crosswalk lines (iv) lane lines

(b) As per the Nagpur plan, the un-surfaced roads were meant for
(i) ODR and village roads (iii) State highway
(ii) Major district road (iv) National highway

(c) MORTH stands for
(i) Ministry of Rail Transport and Highways (iii) Ministry of Road Traffic and Highways
(ii) Ministry of Road Transport and Highways (iv) Ministry of Road, Terrain and Highways

(d) Highway facilities are designed for
(i) annual average hourly volume (iii) thirtieth highest hourly volume
(ii) annual average daily traffic (iv) peak hourly volume of the year

(e) In CBR test the value of CBR is calculated at
(i) 2.5 mm penetration only (iii) 5.0 mm penetration only
(ii) 7.5 mm penetration only (iv) Both 2.5mm and 5.0mm penetration

(f) If aggregate impact value is 20-30 percent, then it is classified as
(i) exceptionally strong (iii) satisfactory for road surfacing
(ii) Strong (iv) unsuitable for road surfacing

(g) Tie bars in cement concrete pavements are at
(i) expansion joints (iii) warping joints
(ii) contraction joints (iv) longitudinal joints

(h) As per IRC recommendations, the maximum limit of superelevation for mixed traffic in plain terrain is
(i) 1 in 15 (iii) 1 in 10
(ii) 1 in 12.5 (iv) Equal to camber

(i) Rigidity factor for a tyre pressure greater than 7kg/cm^2 is
(i) Equal to 1 (iii) Greater than 1
(ii) Less than 1 (iv) Zero

(j) Desire Lines are plotted in
(i) traffic volume studies (iii) accident studies
(ii) speed studies (iv) origin and destination studies

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Q.2 Solve both questions : **[14]**

(a) Define highway alignment. What are the factors affecting highway alignment?

(b) Briefly outline the various road patterns commonly in use.

Q.3 Solve both questions : **[14]**

(a) What is camber and why it is to be provided for a road surface? Explain different types of camber in pavement with suitability of each as per traffic type.

(b) Calculate the length of transition curve using the following data: Design speed = 70 kmph, Radius of circular curve = 200m, Pavement width including extra widening = 7.5m, Allowable rate of introduction of superelevation (pavement rotated about the centre line) 1 in 150.

Q.4 Solve both questions : **[14]**

(a) Explain briefly about various factors which affect the:
(i) Road User Characteristics
(ii) Vehicular Characteristics

(b) What is traffic volume? Enumerate the different methods of carrying out traffic volume studies.

Q.5 Solve both questions : **[14]**

(a) Calculate the length of stopping sight distance for two way traffic in a single lane road. The design speed is 80 kmph. Coefficient of friction is 0.6. Calculate for sloping road with (i) Ascending slope of 2% (ii) Descending slope of 3%.

(b) Describe briefly about PIEV theory.

Q.6 Solve both questions : **[14]**

(a) The radius of a horizontal circular curve is 100m. The design speed is 50 kmph and the design Coefficient of lateral friction is 0.15. Calculate the superelevation required if full lateral friction is assumed to develop.

(b) Explain briefly about different types of transition curves commonly adopted.

Q.7 Solve this question : **[14]**

Define the following terms:
(i) Space-mean speed
(ii) Time-mean speed
(iii) Traffic Island
(iv) Passenger car unit (PCU)
(v) Traffic capacity
(vi) 30th highest hourly volume
(vii) Jam Density

Q.8 Solve both questions : **[14]**

(a) What are the different types of bituminous materials used in road construction? Under what circumstances each of these materials are preferred?

(b) Differentiate between flexible pavements and rigid pavements clearly outlining the advantages and disadvantages of both.

Q.9 Write short notes on any two of the following:- **[14]**

(a) Joints in Cement Concrete pavements

(b) Prime Coat and Tack Coat

(c) Extra Widening in Horizontal Curves

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(d) Floating Car Method of Speed Delay Study

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